Journal of Electrical Engineering & Technology (2026) 21:189–202
https://doi.org/10.1007/s42835-025-02445-6

ORIGINAL ARTICLE

Research on Gain Allocation Strategy of Multi-park Integrated Energy System Based on Cooperative Game and Improved Shapley Value Method

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Received: 19 November 2024 / Revised: 30 July 2025 / Accepted: 8 September 2025 / Published online: 1 October 2025
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Abstract
In the future, an increasing number of integrated energy systems in parks are expected to participate in cooperative games to leverage transactional collaboration among multiple stakeholders and the interaction between various energy sources. Consequently, it is essential to fairly distribute the cooperative benefits of multi-park integrated energy systems. This paper introduces a gain allocation strategy for multi-park integrated energy systems based on cooperative games and an enhanced Shapley value method. The strategy involves establishing a cooperative game model for multi-park integrated energy systems with single-park integrated energy systems as participants. By considering comprehensive costs including energy purchase, operation and maintenance, and transmission loss, the traditional Shapley value method is improved, adhering to principles of overall and individual rationality. The improved Shapley value not only takes into account the contributions of each participant but also includes the costs incurred and risks borne by the participants. This makes the benefit distribution fairer and more realistic. Additionally, it is more scalable and can be applied in more complex practical scenarios. Through case analysis, it was found that the improved Shapley value method reallocates the benefits of the integrated energy systems across various parks. The improved Shapley value method reduced the costs for residential park by 5.86%, increased the costs for commercial park by 3.22%, and raised the costs for industrial park by 4.34%. This has achieved a synergistic effect among the entities, improved resource utilization, and made the benefit distribution within the alliance more rational and balanced.

Keywords Multi-park integrated energy system · Cooperative games · Composite indicators · Improved Shapley values · Distribution of benefits

Symbol Explanation

$P_{P,CHP}$	Electric output power of the CHP units
$P_{E,CHP}$	Thermal output power of the CHP units
μ_{CHP}^{H}	Thermal conversion efficiency of CHP units
μ_{CHP}^{E}	Electrical conversion efficiency of CHP units
L_{NG}	Low calorific value of natural gas
$P_{Q,CHP}$	Gas consumption of CHP units
$P_{C,EC}$	Refrigeration power of the electric refrigerator
COP_{EC}	Energy efficiency coefficient of the electric refrigerator
$P_{E,EC}$	Input electrical power of the electric chiller
$P_{C,AC}$	Refrigeration power of the absorption chiller
μ_{AC}	Refrigeration efficiency of the absorption chiller
$P_{H,AC,in}$	Input thermal power of the absorption chiller

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W_{CS}	Energy stored by the energy storage device
$P_{c,CS}$, $P_{f,CS}$	Charging and discharging power
$C_{buy,i}$	Cost of purchasing energy from outside
$C_{om,i}$	Equipment operation and maintenance cost
$C_{loss,i,j}$	Cost of energy loss between park i and park j
$\mu_{E,t}$ $\mu_{H,t}$ $\mu_{Q,t}$	Prices of electricity, heat, and gas at time t
$P_{E,i,t}$, $P_{H,i,t}$, $P_{Q,i,t}$	Outward power of heat purchase, heat purchase, and gas purchase
k_{PV}	Operation and maintenance costs of PV
k_{WT}	Operation and maintenance costs of WT
k_{CHP}	Operation and maintenance costs of CHP
k_{EC}	Operation and maintenance costs of EC
k_{AC}	Operation and maintenance costs of AC
k_C	Operation and maintenance costs of energy storage
$P_{PV,i,t}$	Output of PV at time t
$P_{WT,i,t}$	Output of WT at time t
$P_{CHP,i,t}$	Output of CHP at time t
$P_{EC,i,t}$	Output of EC at time t
$P_{AC,i,t}$	Output of AC at time t
$P_{c,i,t}$, $P_{f,i,t}$	Charging and discharging power of the energy storage
$P_{EL,i,t}$, $P_{QL,i,t}$, $P_{HL,i,t}$, $P_{CL,i,t}$	Electrical, gas, heat, and cooling load power of park i at time t
α	Boolean functions
$P_{E,CHP,i,\min}$, $P_{C,AC,i,\min}$, $P_{C,EC,i,\min}$	The lower limits of the output of CHP, AC, and EC
$P_{E,CHP,i,\max}$, $P_{C,AC,i,\max}$, $P_{C,EC,i,\max}$	The upper limit of CHP, AC, and EC
$P_{E,i,\min}$, $P_{H,i,\min}$, $P_{Q,i,\min}$	The lower limits of electricity, heat and gas
$P_{E,i,\max}$, $P_{H,i,\max}$, $P_{Q,i,\max}$	The upper limits of electricity, heat and gas
$P_{CS,c,i,\min}$, $P_{CS,f,i,\min}$	The minimum power when charging and discharging the energy storage device
$P_{CS,c,i,\max}$, $P_{CS,f,i,\max}$	The maximum power when charging and discharging the energy storage device
$P_{E,ij,\max}$, $P_{Q,ij,\max}$, $P_{H,ij,\max}$	The maximum transmission power of electricity, gas and heat
x'_{ij}	The normalized form of the indicator matrix X_{ij}
e_j	The entropy value of the j-th indicator

1 Introduction

With the increasing demand for energy and the changing energy landscape [1, 2], integrated energy systems in parks have gained attention for their effective utilization of various energy sources, including electricity, heat, gas, and cold [3]. As the number of park-integrated energy systems grows in the same area, individual park systems are facing challenges in optimizing the use of flexible resources during operation. This has led to a growing trend towards collaborative optimization among multi-park integrated energy systems. These systems [4], which integrate a variety of energy resources, conversion technologies, and intelligent control optimization, offer advantages such as low carbon emissions [5–8], high efficiency, and cost-effectiveness [9–11]. In this context, it is essential to devise a fair and balanced mechanism for distributing comprehensive interests that safeguards the interests of each individual park integrated energy system while ensuring effective energy supply and consumption [12–15]. Cyber security is a key issue in the energy system of the cooperative park, especially when multiple agents and blockchain technology are used to optimize energy distribution and transactions, ensuring the integrity of data, strengthening the protection of the blockchain network, regularly detecting system vulnerabilities, and enhancing the authentication mechanism are the key measures to ensure the security of the energy system of the cooperative park [16–19].

In the multi-park integrated energy system, it is crucial to ensure the self-interest of each park's integrated energy system while constructing a reasonable and balanced distribution mechanism for the overall benefits. In reference [20], An industrial park adopts a performance-based contract energy management model, where the energy service company bears 40% of the technical investment but only

receives 25% of the benefits. Meanwhile, after the park users save 30% on their electricity costs, they do not reinvest in the investors, leading to a significant decline in long-term cooperation enthusiasm. In reference [21], An ecological park adopts the traditional Shapley value allocation method, where the company responsible for environmental technology research and development contributes 30% of the investment but only receives 20% of the benefits. Meanwhile, low-risk entities, leveraging their scale advantages, obtain excessive allocations, triggering a crisis in the stability of the alliance.

Game theory offers an advanced method for addressing the optimization decisions of individuals with diverse interests. It has the capability to integrate energy demand within the integrated energy system of multiple parks, facilitating energy sharing and exchange among these parks. This ultimately helps to coordinate and optimize the stakeholders of the integrated energy system of various parks [22].

Nucleus method and bargaining theory are common methods for the distribution of benefits in cooperative games, but compared with the improved Shapley value method, the nucleus method cannot fully reflect the contributions of all participants in the case of complex cooperation. The allocation algorithm of the bargaining theory has great subjectivity, which cannot ensure the fairness of distribution, while the improved Shapley value method considers all the participation sequences, evaluates the contribution of each participant, and considers the energy purchase, operation and maintenance and loss to determine the comprehensive index for benefit redistribution, which has stronger adaptability and rationality.

Research on multi-agent energy system collaboration and multi-park integrated energy system focuses on optimization and cooperation among individuals. Compared to frameworks such as contract theory, auction mechanisms, and reinforcement learning combined with game theory used for multi-agent energy system collaboration [23–27], the improved Shapley value method does not involve various contract constraints like contract theory. It also addresses the issue that auction mechanisms cannot handle complex cost allocation problems. Furthermore, compared to reinforcement learning, which requires a large amount of data and time for training to perform dynamic decision-making, this method offers higher efficiency and ensures rationality [23, 28–32].

Game theory encompasses non-cooperative games and cooperative games. In non-cooperative games, a master-slave game model was establish [33] with integrated energy system operators as leaders and load aggregators as followers. Additionally, a two-stage optimal scheduling model for an integrated power-gas energy system based on a Stackelberg game real-time pricing mechanism was developed to mitigate the impact of forecasting errors of wind power and load on unit output planning [34]. However, non-cooperative games typically focus solely on individual rational and egoistic decision-making processes, lacking effective cooperation mechanisms and often achieving only local optimal results. Cooperative games, on the other hand, consider both individual rationality and overall interests, garnering research interest from scholars worldwide. For instance, to achieve synchronous coordination of economy and low carbon in the development of multiple interconnected comprehensive energy systems, a method of low-carbon economic optimization operation of a multi-energy station interconnected comprehensive energy system based on cooperative game was proposed [35]. Moreover, a cooperative optimization operation method for a multi-park integrated energy system was introduced, significantly improving solving speed using Shapley value in cooperative games [36]. In these studies on cooperative games, the Shapley value method is used to distribute benefits according to the contribution degree of each participant to the alliance, although the costfactors of different integrated energy systems are not considered.

This paper presents a gain allocation strategy for a multi-park integrated energy system based on a cooperative game and an improved Shapley value method. The aim is to optimize the energy system comprising a business park, industrial park, and residential park. The improved Shapley value method is utilized to calculate the comprehensive index for income redistribution, taking into consideration energy purchase, operation, and park costs. The objective is to ensure a more reasonable and balanced distribution of benefits among the different parks.

2 Material and Methods

2.1 Cooperative Game Theory

Cooperative games are a significant concept in game theory. They highlight the ability of players to cooperate and form alliances to maximize benefits for the entire group. In these games, it is crucial to establish a fair benefit distribution mechanism to ensure seamless coordination and cooperation within the alliance, ultimately leading to the maximization of benefits.

There are two basic principles of cooperative games:

The principle of overall rationality: for the alliance as a whole, the benefits achieved under the specified operational mechanisms of the cooperative alliance exceed the cumulative benefits that would be realized if the individual members operated separately.

The principle of individual rationality: for the members of the alliance who participate in the cooperation, the benefits obtained according to the benefit distribution mechanism specified by the alliance are greater than the benefits obtained by the alliance members when they operate alone.

2.2 Multi-park Integrated Energy System Cooperation Game Architecture

Considering the energy mutual benefit relationship between parks, this paper presents the cooperative game architecture of multi-park integrated energy system as shown in Fig. 1.

The members of the cooperative alliance are the integrated energy systems of each single park. There are two ways of energy interaction between the members of the alliance and the outside world: 1) Energy exchange among the integrated energy systems of every single park in the alliance; 2) The integrated energy system of each park purchases energy from the power grid, heat network and gas.

2.3 Device Model

2.3.1 Combined Heat and Power Machine

$$\begin{cases} P_{H,CHP} = \mu_{CHP}^{H} L_{NG} P_{Q,CHP} \\ P_{E,CHP} = \mu_{CHP}^{E} L_{NG} P_{Q,CHP} \end{cases} \tag{1}$$

In the context of combined heat and power (CHP) units, $P_{P,CHP}$ and $P_{E,CHP}$ represent the electrical and thermal output power, respectively. The terms μ_{CHP}^{H} and μ_{CHP}^{H} denote the electrical and thermal conversion efficiencies of the CHP units. Additionally, L_{NG} refers to the low calorific value of natural gas, while $P_{Q,CHP}$ indicates the gas consumption of the CHP units.

2.3.2 Electric Refrigerator

$$P_{C,EC} = COP_{EC} P_{E,EC} \tag{2}$$

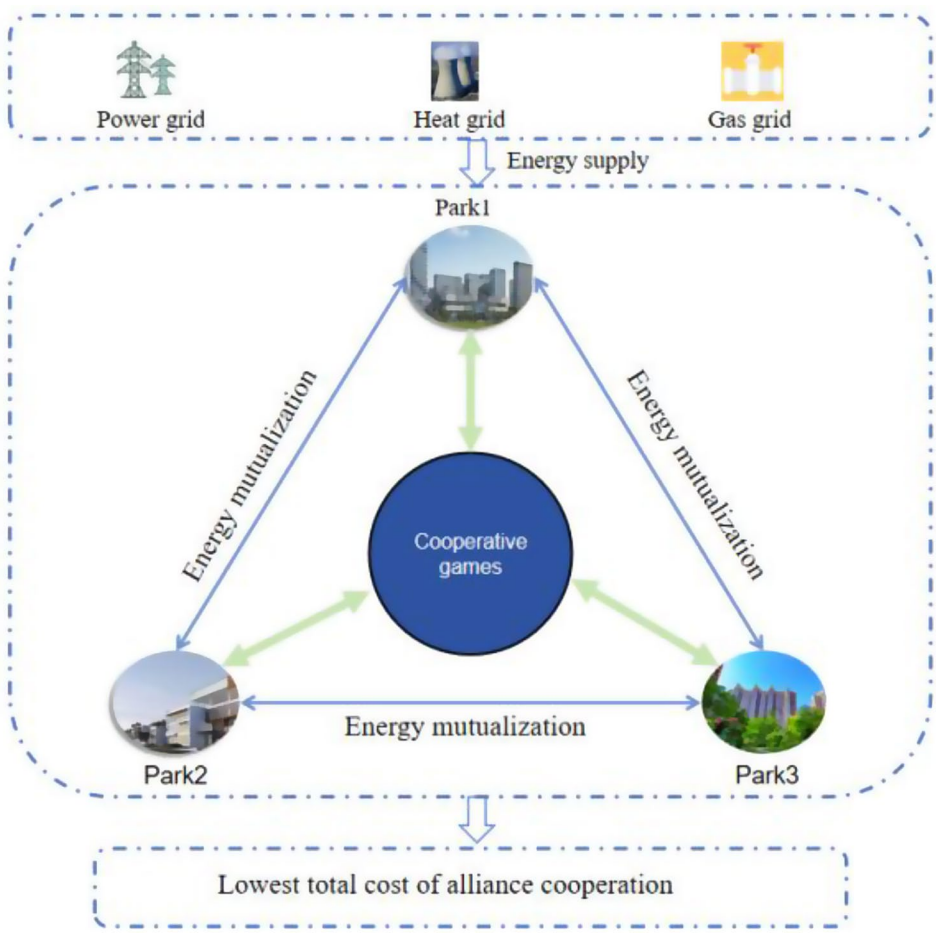

Fig. 1 Diagram of the cooperative game of multi-park integrated energy system

Where $P_{C,EC}$ is the refrigeration power of the electric refrigerator; COP_{EC} is the energy efficiency coefficient of the equipment; $P_{E,EC}$ is the input electrical power of the electric chiller.

2.3.3 Absorption Chiller

$$P_{C,AC} = \mu_{AC} P_{H,AC,in} \tag{3}$$

Where $P_{C,AC}$ is the refrigeration power of the absorption chiller; μ_{AC} is the refrigeration efficiency of the absorption chiller; $P_{H,AC,in}$ is the input thermal power of the absorption chiller.

2.3.4 Energy Storage Equipment

$$W_{CS}(t) = W_{CS}(t-1) + [\mu_c P_{c,CS}(t) - \frac{P_{f,CS}(t)}{\mu_f}]\Delta t \tag{4}$$

Where W_{CS} is the energy stored by the energy storage device; $P_{c,CS}$ and $P_{f,CS}$ represent the charging and discharging power. μ_c and μ_f represent the loss coefficients of charging and discharging processes, and Δt indicates the time interval.

2.4 Single-campus Integrated Energy System Model

2.4.1 Objective Function of a Single-park Integrated Energy System

For a single-park integrated energy system C_i in the non-cooperative game mode, the objective function can be expressed as:

$$\min C_i = \min(C_{buy,i} + C_{om,i}) \tag{5}$$

Where $C_{buy,i}$ is the cost of purchasing energy from outside (including electricity, heat, and gas) C_i is the integrated energy system of a park, $C_{om,i}$ is the equipment operation and maintenance cost of the integrated energy system C_i of a park (including wind turbines, photovoltaics, CHP, absorption chillers, energy storage equipment, etc.), and the expressions between $C_{buy,i}$ and $C_{om,i}$ are as follows:

$$C_{buy,i} = \sum_{i=1}^{T} (\mu_{E,t} P_{E,i,t} + \mu_{H,t} P_{H,i,t} + \mu_{Q,t} P_{Q,i,t}) \tag{6}$$

$$\begin{aligned} C_{om,i} = \sum_{i=1}^{T} (&k_{PV} P_{PV,i,t} + k_{WT} P_{WT,i,t} + k_{CHP} P_{CHP,i,t} \\ &+ k_{EC} P_{EC,i,t} + k_{AC} P_{AC,i,t} + k_C (P_{c,i,t} + P_{f,i,t})) \end{aligned} \tag{7}$$

Where $\mu_{E,t}$, $\mu_{H,t}$, and $\mu_{Q,t}$ are the prices of electricity, heat, and gas at time t; T is the scheduling period, and this paper takes 24 hours; $P_{E,i,t}$, $P_{H,i,t}$, and $P_{Q,i,t}$ are the outward power of heat purchase, heat purchase, and gas purchase in the park; k_{PV}, k_{WT}, k_{CHP}, k_{EC}, k_{AC}, and k_C represent the unit operation and maintenance costs of PV, WT, CHP, EC, AC and energy storage equipmen, and the $P_{PV,i,t}$, $P_{WT,i,t}$, $P_{CHP,i,t}$, $P_{EC,i,t}$, and $P_{AC,i,t}$ are the output of PV, WT, CHP, AC and EC equipment at time t. $P_{c,i,t}$ and $P_{f,i,t}$ respectively represent the charging and discharging power of the energy storage equipment at time t, and the charging and discharging of energy storage cannot be carried out at the same time.

2.4.2 Constraints of a Single-park Integrated Energy System

(1) Power Balance Constraints

$$\begin{cases} P_{E,i,t} + P_{WT,i,t} + P_{PV,i,t} + P_{E,CHP,i,t} + \alpha_{ES,f,i,t} P_{ES,f,i,t} \\ = P_{E,EC,i,t} + P_{EL,i,t} + \alpha_{ES,c,i,t} P_{ES,c,i,t} \\ P_{Q,i,t} + \alpha_{QS,f,i,t} P_{QS,f,i,t} \\ = P_{QL,i,t} + P_{Q,CHP,i,t} + \alpha_{QS,c,i,t} P_{QS,c,i,t} \\ P_{H,i,t} + P_{H,CHP,i,t} + \alpha_{HS,f,i,t} P_{HS,f,i,t} \\ = P_{HL,i,t} + P_{H,AC,i,t} + \alpha_{HS,c,i,t} P_{HS,c,i,t} \\ P_{C,AC,i,t} + P_{C,EC,i,t} + \alpha_{CS,f,i,t} P_{CS,f,i,t} \\ = P_{CL,i,t} + \alpha_{CS,c,i,t} P_{CS,c,i,t} \end{cases} \tag{8}$$

Where $P_{EL,i,t}$, $P_{QL,i,t}$, $P_{HL,i,t}$, $P_{CL,i,t}$ are the electrical, gas, heat, and cooling load power of park i at time t; $\alpha_{ES,c,i,t}$, $\alpha_{ES,f,i,t}$, $\alpha_{QS,c,i,t}$, $\alpha_{QS,f,i,t}$, $\alpha_{HS,c,i,t}$, $\alpha_{HS,f,i,t}$, $\alpha_{CS,c,i,t}$, $\alpha_{CS,f,i,t}$ are Boolean functions, and their purpose is to ensure that the charging and discharging of energy storage can not be carried out at the same time.

(2) Equipment Output Constraints

$$\begin{cases} P_{E,CHP,i,\min} \le P_{E,CHP,i,t} \le P_{E,CHP,i,\max} \\ P_{C,AC,i,\min} \le P_{C,AC,i,t} \le P_{C,AC,i,\max} \\ P_{C,EC,i,\min} \le P_{C,EC,i,t} \le P_{C,EC,i,\max} \end{cases} \tag{9}$$

Where $P_{E,CHP,i,\min}$, $P_{C,AC,i,\min}$, $P_{C,EC,i,\min}$ are the lower limits of the output of CHP, AC, and EC; $P_{E,CHP,i,\max}$, $P_{C,AC,i,\max}$, $P_{C,EC,i,\max}$ are the upper limits of CHP, AC, and EC.

(3) Constraints on Energy Purchase

$$\begin{cases} P_{E,i,\min} \le P_{E,i,t} \le P_{E,i,\max} \\ P_{H,i,\min} \le P_{H,i,t} \le P_{H,i,\max} \\ P_{Q,i,\min} \le P_{Q,i,t} \le P_{Q,i,\max} \end{cases} \tag{10}$$

Where $P_{E,i,\min}$, $P_{H,i,\min}$, $P_{Q,i,\min}$ represent the lower limits of electricity, heat and gas; $P_{E,i,\max}$, $P_{H,i,\max}$, $P_{Q,i,\max}$ represent the upper limits of electricity, heat and gas.

(4) Conditional Constraints on Energy Storage Equipment

$$\begin{cases} P_{CS,c,i,\min} \le P_{CS,c,i,t} \le P_{CS,c,i,\max} \\ P_{CS,f,i,\min} \le P_{CS,f,i,t} \le P_{CS,f,i,\max} \\ \alpha_{CS,f,i,t} \cdot \alpha_{CS,c,i,t} = 0 \\ \alpha_{CS,f,i,t} + \alpha_{CS,c,i,t} = 1 \end{cases} \tag{11}$$

Where $P_{CS,c,i,\min}$, $P_{CS,f,i,\min}$ respectively, is the minimum power when charging and discharging the energy storage device; $P_{CS,c,i,\max}$, $P_{CS,f,i,\max}$ represent the maximum power when charging and discharging the energy storage device.

2.5 Multi-park Integrated Energy System Model

2.5.1 Multi-park Integrated Energy System Objective Function

When multiple park-integrated energy systems jointly participate in the cooperative game to form a cooperative alliance, the objective function of the alliance is the comprehensive cost of the cooperative alliance, which can be expressed as follows:

$$\min C_v = \sum_{i,j=1}^{M} (C_{buy,i} + C_{om,i} + C_{loss,i,j}) \tag{12}$$

Where the total cost of the cooperative alliance formed by the integrated energy system of multiple parks consists of the energy purchase cost, operation and maintenance cost and energy transmission loss of the parks participating in the cooperative game; M denotes the number of parks participating in the cooperative game; and the cost of energy loss $C_{loss,i,j}$ due to energy mutualization between park i and park j can be expressed as follows:

$$C_{loss,i,j} = \frac{1}{2}\sum_{t=1}^{T} (\mu_{E,t}\delta_E P_{E,ij,t} + \mu_{H,t}\delta_H P_{H,ij,t} + \mu_{Q,t}\delta_Q P_{Q,ij,t}) \tag{13}$$

Where δ_E, δ_H, δ_Q represent the transmission loss coefficients of electrical energy, thermal energy, and gas energy for energy mutualization.

2.5.2 Multi-campus Integrated Energy System Constraints

Parks participating in the cooperative game should likewise satisfy the equipment output constraints, energy purchase constraints, and energy storage constraints, which are described in Section 1.4.2.

(1) Power Balance Constraints

The power balance constraint when park i participates in the cooperative game should consider the energy mutualization between parks as follows:

$$\begin{cases} P_{E,i,t} + P_{WT,i,t} + P_{PV,i,t} + P_{E,CHP,t} + \alpha_{ES,f,i,t}P_{ES,f,i,t} \\ = P_{E,EC,i,t} + P_{EL,i,t} + \alpha_{ES,c,i,t}P_{ES,c,i,t} + P_{E,ij,t} \\ P_{Q,i,t} + \alpha_{QS,f,i,t}P_{QS,f,i,t} \\ = P_{QL,i,t} + P_{Q,CHP,i,t} + \alpha_{QS,c,i,t}P_{QS,c,i,t} + P_{Q,ij,t} \\ P_{H,i,t} + P_{H,CHP,i,t} + \alpha_{HS,f,i,t}P_{HS,f,i,t} \\ = P_{HL,i,t} + P_{H,AC,i,t} + \alpha_{HS,c,i,t}P_{HS,c,i,t} + P_{H,ij,t} \\ P_{C,AC,i,t} + P_{C,EC,i,t} + \alpha_{CS,f,i,t}P_{CS,f,i,t} \\ = P_{CL,i,t} + \alpha_{CS,c,i,t}P_{CS,c,i,t} \end{cases} \tag{14}$$

(2) Energy Transfer Constraints

$$\begin{cases} -P_{E,ij,\max} \le P_{E,ij,t} \le P_{E,ij,t,\max} \\ -P_{Q,ij,\max} \le P_{Q,ij,t} \le P_{Q,ij,t,\max} \\ -P_{H,ij,\max} \le P_{H,ij,t} \le P_{H,ij,t,\max} \end{cases} \tag{15}$$

Where $P_{E,ij,\max}$, $P_{Q,ij,\max}$, $P_{H,ij,\max}$ represent the maximum transmission power of electricity, gas and heat, and positive and negative represent the direction of energy transmission.

3 Solving Cooperative Game Model for Multi-park Integrated Energy System

3.1 Shapely Value Method

Shapley value method [37] is a concept in cooperative game theory, which is a strategy to distribute benefits according to the contribution of each participant after a cooperative game, and the method is established based on three axioms: the symmetry axiom, the validity axiom, and the addition axiom. The first axiom explains that the order of stakeholders has no effect on the distribution of benefits. The second axiom explains that the total value of the coalition of stakeholders is the sum of the Shapley values. The third

axiom explains that the assigned values of stakeholders are additive.

The unique indicator that describes the value of each player in the cooperative game B(N,V) in the Shapley value method is the vector ($\varphi_1, \varphi_2, \varphi_3, \ldots \varphi_n$), where:

$$\varphi_i = \frac{(n-k)!(k-1)!}{n!}[V(S) - V(\frac{S}{i})] \tag{16}$$

Where n is the total number of people in the cooperative coalition; k is the number of game parties included in the coalition S; call the vector ($\varphi_1, \varphi_2, \varphi_3, \ldots \varphi_n$) be the Shapley values of the game B(N, V), φ_i is the Shapley value of game party i; it represents the effect of game party i's participation or non-participation in the cooperative game on the coalition's characteristic function, and also responds to game party it's contribution to the coalition S; $\frac{(n-k)!(k-1)!}{n!}$ refers to the probability that game party i participates in coalition S.

Therefore, the Shapley value of each player is the expected contribution value of its participation in the cooperative coalition, which can be used as a measure of the value of each player in the cooperative game.

3.2 Improved Shapley Value Method

The improved Shapley value method proposed in this paper enhances the traditional Shapley value approach by incorporating three key factors: energy procurement costs, operational costs, and loss costs. This establishes a comprehensive index for redistributing benefits, which not only accounts for each participant's contribution but also considers their incurred costs and assumed risks. As a result, the profit allocation becomes fairer, more realistic, and more scalable. The refined method can be applied to more complex real-world scenarios, making it better suited for future market environments in integrated energy systems.

The entropy value method is used to determine the weights, if there are n integrated energy systems and m indicators, the indicator matrix can be expressed as:

$$X_{ij} = \begin{pmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{pmatrix} \tag{17}$$

$$e_j = -(\ln(n))^{-1} \sum_{i=1}^{n} (\frac{x'_{ij}}{\sum\limits_{i=1}^{n} x'_{ij}}) \ln(\frac{x'_{ij}}{\sum\limits_{i=1}^{n} x'_{ij}}) \tag{18}$$

Where x'_{ij} is the normalized form of the indicator matrix X_{ij}; e_j is the entropy value of the j-th indicator. The weight value of the j-th indicator can be expressed as:

$$\omega_i = \frac{g_j}{\sum\limits_{j=1}^{m} g_j} \tag{19}$$

Where g_j = 1-e_j; ω_i takes a value between 0 and 1.

First, the weight coefficient matrix is constructed after confirming the weight values:

$$\omega = [\omega_1 \quad \omega_2 \quad \omega_3]^T \tag{20}$$

The proportion of energy purchase cost is expressed as k_1, the proportion of Operational and Maintenance cost is expressed as k_2, and the proportion of depletion cost is expressed as k_3, with the expression:

$$k_1 = \frac{C_{buy}}{C_{buy,N}} \tag{21}$$

$$k_2 = \frac{C_{om}}{C_{om,N}} \tag{22}$$

$$k_3 = \frac{C_{loss}}{C_{loss,N}} \tag{23}$$

Then, the following indicators are given for business park, industrial park, and residential park: purchased energy indicator k_{c1}, k_{i1}, k_{r1}, operation and maintenance indicator k_{c2}, k_{i2}, k_{r2}, and loss indicator k_{c3}, k_{i3}, k_{r3}. Construct the matrix of correction coefficients B:

$$B = [k_1 \quad k_2 \quad k_3] = \begin{pmatrix} k_{c1} & k_{c2} & k_{c3} \\ k_{i1} & k_{i2} & k_{i3} \\ k_{r1} & k_{r2} & k_{r3} \end{pmatrix} \tag{24}$$

K^* is set as a comprehensive performance index with the expression:

$$K^* = B \cdot \omega = k_1\omega_1 + k_2\omega_2 + k_3\omega_3 = \begin{pmatrix} K_C^* \\ K_I^* \\ K_R^* \end{pmatrix} \tag{25}$$

Where K_C^*, K_I^*, K_R^* represent the comprehensive indicators of business park, industrial park, and residential park.

The modified Shapley value method has a correction coefficient of income distribution C_i, The correction coefficient vector can be expressed as: $K_i^* = [K_1^* \quad K_2^* \quad \cdots \quad K_n^*]$, n is the number of members of the consortium topic. Correction

factors for the integrated energy system of each park ΔC_n, expressed as:

$$\Delta K_i^* = K_i^* - \frac{1}{n} \tag{26}$$

The results of the income redistribution of the integrated energy system of each park based on the improved Shapley value method:

$$\varphi_i^* = \varphi_i + \Delta K_i^* C_{park} \tag{27}$$

The Shapley value method is optimized according to the comprehensive indicators K^* to obtain the modified cost of business park, industrial park, and residential park as:

$$\begin{cases} \varphi_C^* = \varphi_C + (K_C^* - \frac{1}{n})C_{commerce} \\ \varphi_I^* = \varphi_I + (K_I^* - \frac{1}{n})C_{industrial} \\ \varphi_R^* = \varphi_R + (K_R^* - \frac{1}{n})C_{resident} \end{cases} \tag{28}$$

Where φ_C, φ_I, φ_R are the allocation result of the unimproved Shapley value method and n is the number of members in the cooperative union.

3.3 Solution Method

In this paper, the solving tool (Yalmip) in Matalab R2018b is used to solve the constructed cooperative gaming coalition of multi-campus integrated energy systems. The benefits of the cooperative coalition are allocated using the Shapley value method. After that, the benefits of the cooperative coalition are redistributed using the improved Shapley value method proposed in this paper. The solution process is shown in Fig. 2.

4 Calculus Analysis

4.1 Simulation Settings

The paper's example comes from an energy system in eastern China that includes business park, industrial park, and residential park managed by the same energy operator. The business and residential parks' equipment includes fans, photovoltaics, absorption chillers (AC), electric chillers (EC), combined heat and power (CHP) units, and energy storage equipment for electricity, gas, heat, and cold. The loads comprise electric load, gas load, heat load, and cold load. The industrial park's equipment includes fans, photovoltaic (PV), cogeneration units (CHP), and energy storage equipment for electricity, gas, heat, and cold, and its loads consist of electric load, gas load, and heat load. You can

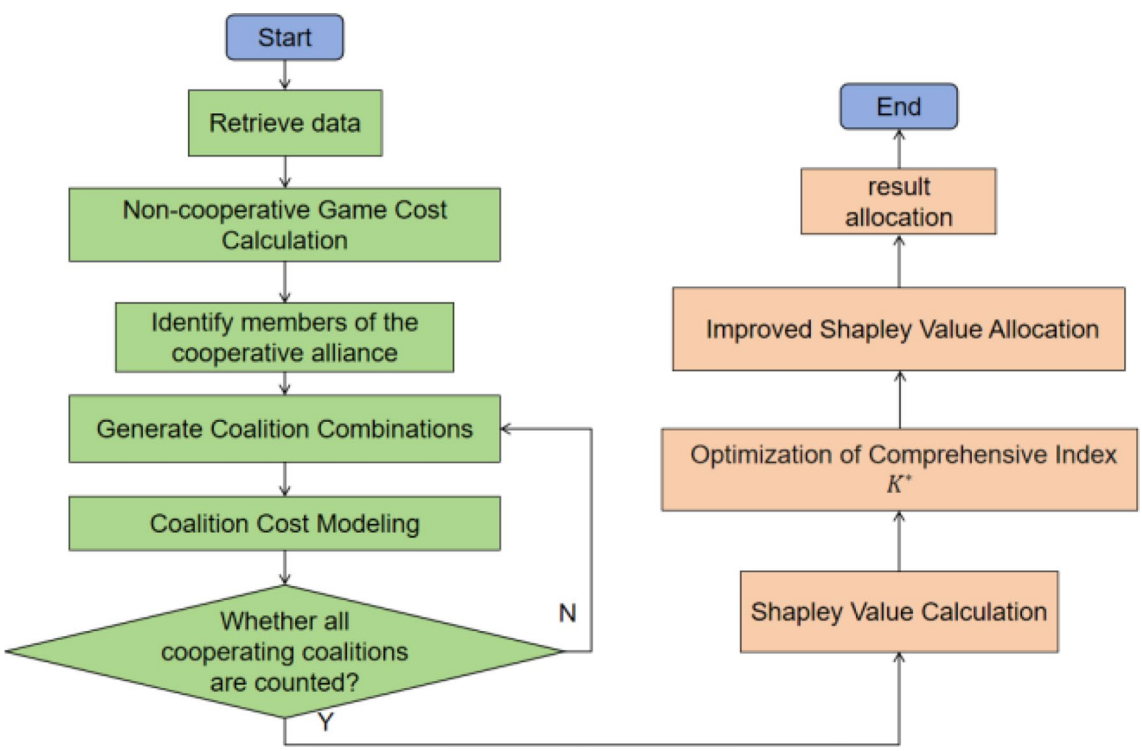

Fig. 2 Flowchart of cooperative game model solving

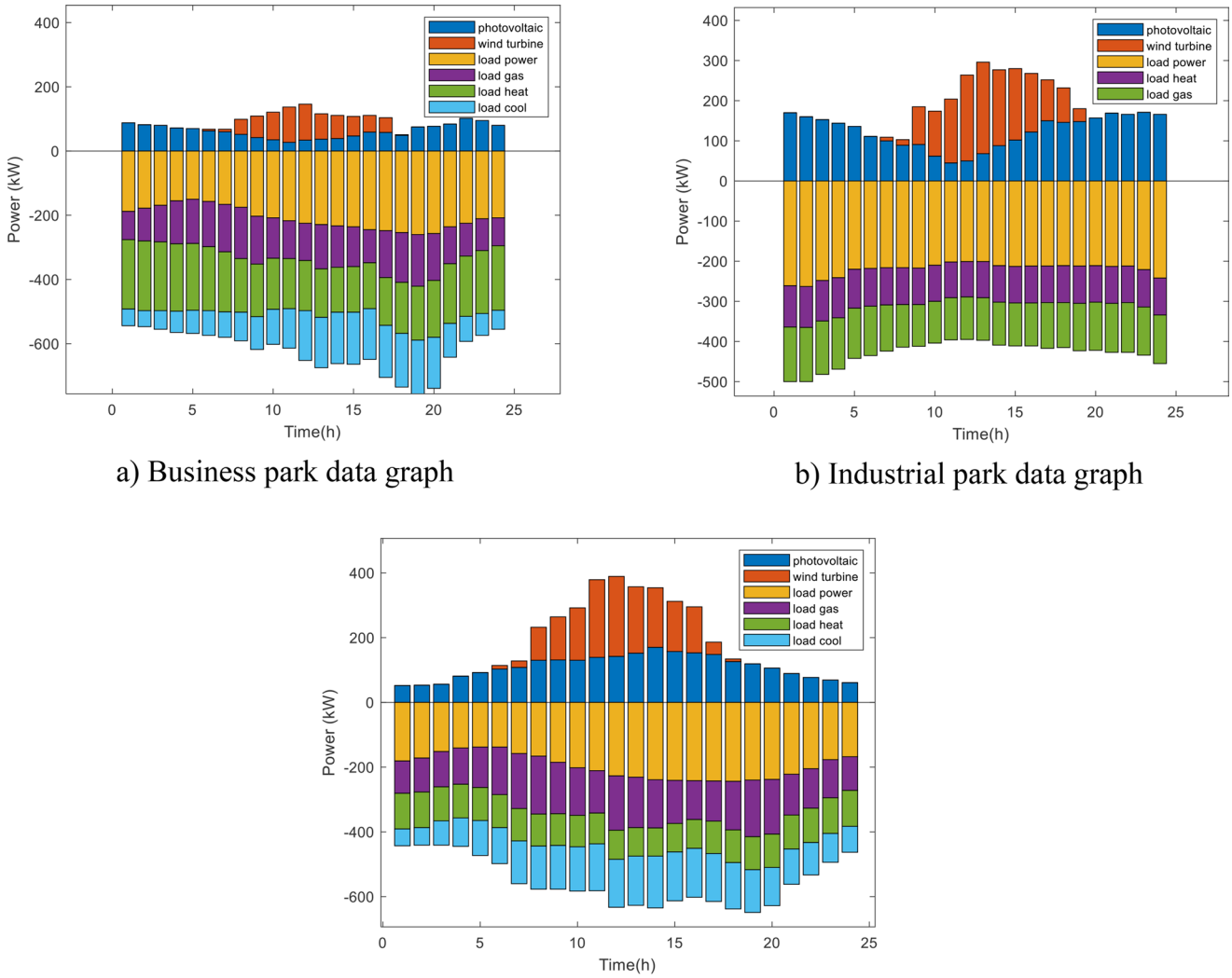

Fig. 3 Wind, solar and load data diagram of the integrated energy system of the three parks

find the data for wind turbines, PV, and each load in the three parks in Fig. 3, and their internal structure is shown in Fig. 4.

There is energy mutualization between the parks, and the transmission losses during transmission are 6, 3 and 4% for electricity, heat and gas, respectively. Some of these parameters (energy prices, operating parameters of the park equipment) are shown in Tables 1 and 2:

4.2 Comparison of Benefit-sharing Programs for Cooperative Alliances

To synergistically optimize a multi-park integrated energy system consisting of business park, industrial park, and residential park, and to allocate benefits to the park costs, the following scenarios are set up for a comparative analysis of the results of benefit allocation.

Scenario 1: There is no energy interaction among multiple parks, and each park operates independently;

Scenario 2: There is energy interaction between parks, the multi-park integrated energy system is coordinated and optimized, and each park is allocated according to its own cost

Scenario 3: Multiple parks form a gaming coalition, and the Shapley value method is used to allocate the costs of each park

Scenario 4: The improved Shapley value method proposed in this paper is used to allocate the costs of each park.

The results of the four scenarios are shown in Table 3 below:

In order to better present the results, the data in the table are visualized and the results are shown in Fig. 5.

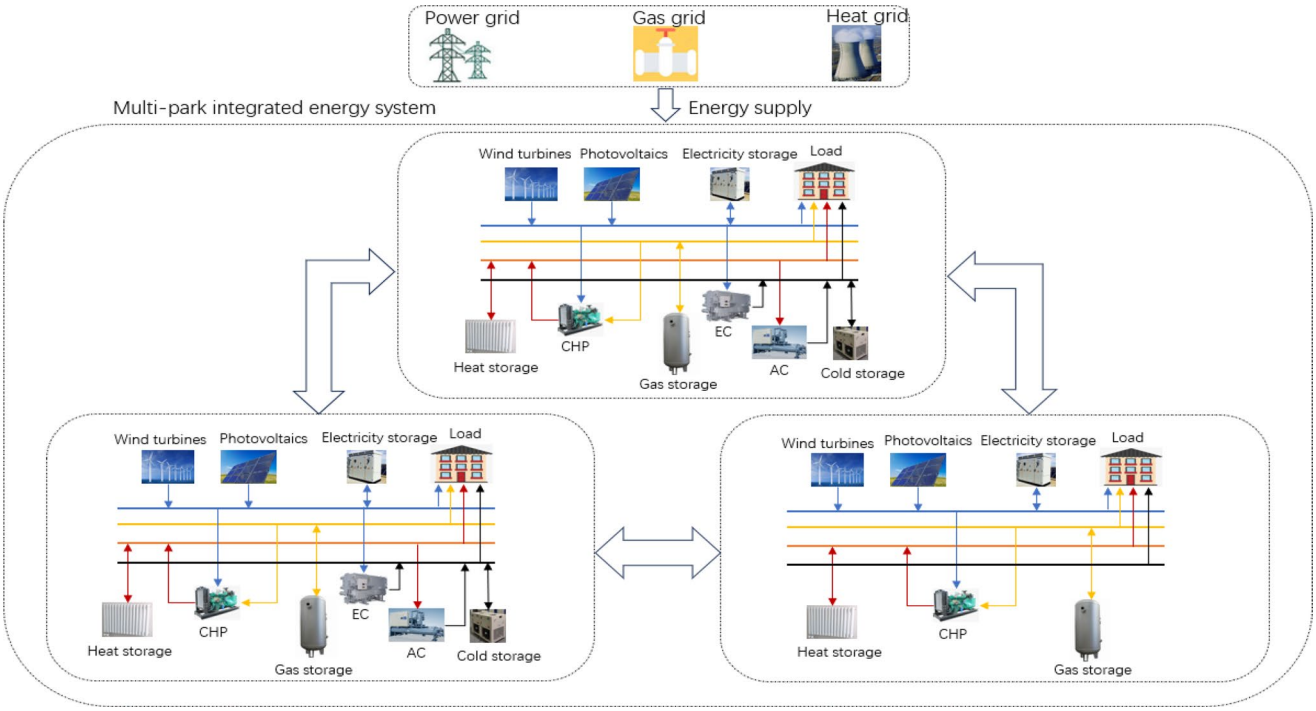

Fig. 4 Structural diagram of a multi-park integrated energy system

Table 1 Time-of-use price of energy

Energy prices（Yuan/kWh）			
Time（h）	Elec price	Heat price	Gas price
1–7,23–24	0.48	0.35	0.25
8–11,15–18	0.88	0.35	0.25
12–14,19–22	1.10	0.35	0.25

Table 2 Business park equipment parameters

Park	Parameter	Numeric value	Parameter	Numeric value
Business park	$P_{E,1,min}$	0 kW	$P_{E,1,max}$	200 kW
	$P_{H,1,min}$	0 kW	$P_{H,1,max}$	200 kW
	$P_{Q,1,min}$	0 kW	$P_{Q,1,max}$	200 kW
	$P_{E,CHP,min}$	0 kW	$P_{E,CHP,max}$	60 kW
	$P_{C,AC,min}$	0 kW	$P_{C,AC,max}$	80 kW
	$P_{C,EC,min}$	0 kW	$P_{C,EC,max}$	100 kW

Table 3 Allocation results in different scenarios

Cost	Business park	Industrial park	Residential park
Scenario 1	262.0848	233.9718	359.9031
Scenario 2	244.8648	285.8921	308.0542
Scenario 3	250.9134	230.4045	357.4933
Scenario 4	258.9934	240.4045	336.5453

As can be seen from Fig. 5, for Scenario 1, the cost is higher compared to the other scenarios because there is no energy mutualization among the parks;

For Scenario 2, energy mutualization among parks is realized by forming cooperative gaming alliance among parks, which can effectively improve the cost;

For Scenario 3, the Shapley value method is allocated based on the contribution of each park, in which the cost of business park is improved by 2.47%, the cost of industrial park is reduced by 19.4%, and the cost of residential park is improved by 16.04%, which ensures fairness, and the allocation results are more reasonable and balanced;

For Scenario 4, the improved Shapley value method proposed in this paper is utilized to determine the comprehensive index by taking into account the park's purchased energy, transportation and loss, and on the basis of the Shapley value method, the cost of residential park has been reduced, and the cost of business park and industrial park has been increased, so that the distribution results are more reasonable.

4.3 Distribution Analysis Based on Improved Shapley Value Methodology

The composite indicators mentioned above can influence the effectiveness of the enhanced Shapley value method in different ways depending on the specific circumstances. By the improved Shapley value theory, we derived a matrix of correction coefficients (B) and a matrix of weighting coefficients (ω) as follows:

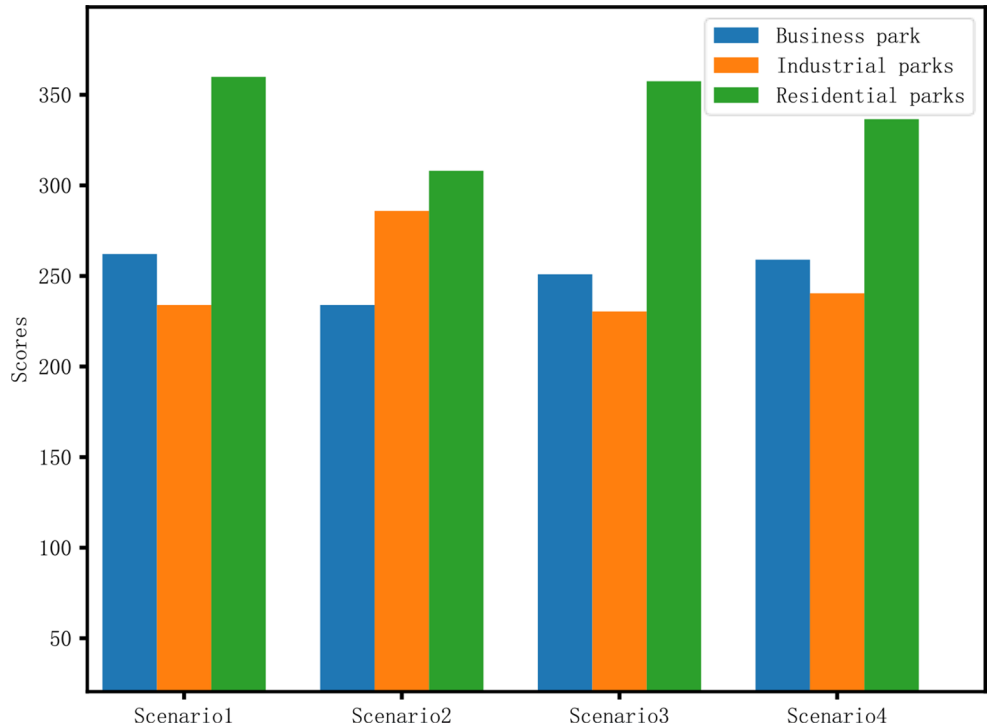

Fig. 5 Cost allocation results for four scenarios

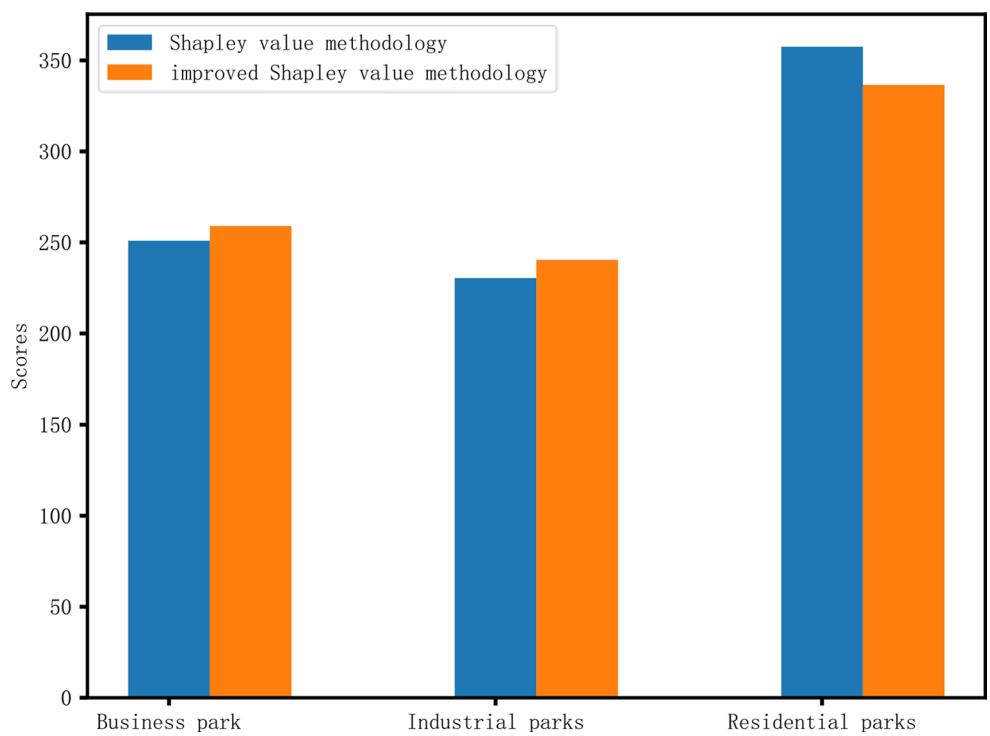

Fig. 6 Comparison of the results before and after the improvement of the Shapley value method

$$B = \begin{bmatrix} 1.019 & 0.6771.423 \\ 0.91 & 1.1761.233 \\ 1.072 & 1.1470.357 \end{bmatrix} \tag{29}$$

$$\omega = [0.3357 \quad 0.2526 \quad 0.4117]^T \tag{30}$$

D is obtained by knowing the matrix of correction coefficients ω and the matrix of weighting coefficients K^*:

$$K^* = \begin{pmatrix} K_C^* \\ K_I^* \\ K_R^* \end{pmatrix} = \begin{bmatrix} 0.366 \\ 0.368 \\ 0.265 \end{bmatrix} \tag{31}$$

The allocation results of the improved Shapley value method can be obtained by calculating Eq. (26), and the comparison of the results of the improved Shapley value method and the pre-improved Shapley value method is shown in Fig. 6:

According to Fig. 6, it can be seen that the improved Shapley value method reduces the cost of residential park and raises the cost of business park and industrial park, and compared with the Shapley value method, the improved Shapley value method reduces the cost of residential park by 5.86%, raises the cost of business park by 3.22%, and raises the cost of industrial park by 4.34%. This is because the improved Shapley value method takes into account comprehensive indicators of commercial, industrial, and residential parks. The comprehensive indicators are determined based on the proportion of energy purchasing, operation and maintenance, and losses of the three parks within the entire cooperative alliance. The improved Shapley value method considers factors that can reduce transportation losses or optimize operation and maintenance, thereby lowering overall costs. The parks will receive corresponding cost distribution shares, rather than simply sharing costs according to a fixed ratio. This distribution method achieves efficiency (minimizing total costs) while ensuring fairness.

5 Conclusion

In this paper, the research on the cooperative and optimal operation of multi-park integrated energy systems as well as the rational allocation of the cooperative game is carried out, and the following conclusions are obtained:

The cooperative game model considering the energy mutual aid among parks is constructed by forming a cooperative alliance of multi-park integrated energy systems with each single park integrated energy system as a participant, with the objective of minimizing the total cost of the alliance. The cooperative alliance composed of business park, industrial park and residential park has a great improvement in economic efficiency compared with the operation of each park alone, and the Shapley value method is used to distribute the benefits of the alliance in a more reasonable way, so that the cost of business park increases by 2.47%, the cost of industrial park decreases by 19.4%, and the cost of residential park increases by 16.04%.

This paper utilizes the improved Shapley value method, which takes into account the purchased energy, transportation and loss components of the park costs to determine a comprehensive index to redistribute the benefits, to redistribute the costs of the three parks, and the improved Shapley value method reduces the costs of the residential park by 5.86%, raises the costs of the business park by 3.22%, and raises the costs of the industrial park by 4.34%. By applying the improved Shapley value method can make the distribution more balanced within the multi-park integrated energy system cooperative alliance.

The integrated energy system is a dynamic and complex system, influenced by various factors such as policy, market fluctuations, and technological advancements. Additionally, differences in energy demand, resource endowments, technological levels, and policy environments between different parks may lead to limitations in the application of the model. Future research may need to account for these long-term dynamic changes and could consider developing more complex multi-level benefit distribution models to further refine the roles and contributions of various stakeholders within the parks, thereby improving the accuracy and applicability of the model.

Author Contributions Xiyun Yang: Conceptualization, Methodology, Project administration, Supervision, Writing – Review & Editing. Zhangzheng Liu: Conceptualization, Methodology, Visualization, Project administration, Investigation, Writing – Original Draft. He Chen: Investigation, Project administration. Lei Yang: Investigation, Project administration.

Funding No funding was received for conducting this study.

Data Availability The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Declarations

Competing Interests The authors declare that they have no competing interests.

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